Coverage Without Citizenship: Synthetic-Control Evidence from California's 2024 Medi-Cal Adult Expansion

Abstract

Background: California's Ages 26 through 49 Adult Full-Scope Medi-Cal Expansion expanded full-benefit Medicaid-like coverage to income-eligible residents regardless of immigration status. Evidence on these state-funded immigrant coverage expansions remains limited because most public surveys identify noncitizens rather than undocumented residents.

Methods: This analysis constructs a 2018-2024 ACS state-year panel for low-income noncitizens ages 26-49. The first-pass analysis compares California with a synthetic donor state constructed from states without comparable adult immigrant coverage policies. Outcomes are uninsurance, Medicaid or means-tested public coverage, any public coverage, and private coverage. The primary estimates use ridge-augmented synthetic control and are interpreted as an activation screen.

Results: In the post-policy period, uninsurance was 26.8 percent in California, compared with an ASCM counterfactual of 34.3 percent, a gap of -7.5 percentage points. Medicaid or means-tested public coverage was 54.9 percent, compared with a counterfactual of 43.1 percent, a gap of +11.8 percentage points.

Conclusions: California shows a directionally promising coverage response, with lower uninsurance and higher Medicaid coverage in the exposed ACS target cell. The placebo rank statistic is not strong enough to treat this as a final causal estimate, so the next pass should focus on diagnostics and estimator comparison.

Introduction

States have increasingly used state funds to close eligibility gaps left by federal Medicaid rules for noncitizens. California's Ages 26 through 49 Adult Full-Scope Medi-Cal Expansion is a particularly useful policy case because it created a discrete eligibility change for a clearly defined low-income noncitizen target population. The policy question is not only whether state-funded expansions increase enrollment, but whether they measurably reduce uninsurance in population survey data that are available quickly enough for policy evaluation.

This paper asks whether California's Ages 26 through 49 Adult Full-Scope Medi-Cal Expansion, effective January 1, 2024, changed coverage among low-income noncitizens ages 26-49. The analysis is deliberately framed around the public-data estimand: ACS noncitizens in the exposed age and income cell. That target does not perfectly identify undocumented residents, but it captures the population group most likely to be moved by the policy.

This study provides an early public-data evaluation of the coverage response to a state-funded immigrant coverage expansion. It does not replace the remaining design work: the next version should audit the donor policy calendar, compare ASCM with SDID and generalized synthetic control, and add access or utilization outcomes if public survey data can support them.

Methods

Data and Population

The analysis uses the IPUMS ACS 1-year extract for 2018-2024. The sample retains household residents who are noncitizens, have family income from 1 to 200 percent of the federal poverty line, and fall in the exposed age band: ages 26-49. Person-level records are collapsed to state-year cells using ACS person weights.

The primary outcomes are uninsurance, Medicaid or means-tested public coverage, any public coverage, and private coverage. Insurance variables are harmonized from the staged IPUMS extract, which stores insurance indicators as binary 0/1 fields.

Policy Timing

The treatment is California's Ages 26 through 49 Adult Full-Scope Medi-Cal Expansion, effective January 1, 2024. The primary post-treatment window is 2024. The donor pool excludes states with comparable adult immigrant-coverage policies during the analysis window or other policy overlap.

Statistical Analysis

For each outcome, the analysis estimates a state-level synthetic-control model and a ridge-augmented synthetic-control model. The donor weights minimize pre-treatment differences between California and a convex combination of donor states. The augmented model then uses ridge regression on donor pre-treatment histories to partially correct interpolation bias when California is not exactly in the donor convex hull. The tables report treated observed values, ASCM counterfactuals, gaps in percentage points, pre-treatment RMSPE, and a placebo-in-space rank statistic for uninsurance.

Results

Table 1 describes the target-population coverage panel. Figure 1 shows raw pre-policy coverage trends for California and the donor-pool mean. In the post-policy period, the ASCM estimates show lower uninsurance and higher Medicaid coverage in the treated state relative to its synthetic counterfactual.

For uninsurance, the observed post-period value was 26.8 percent, compared with an ASCM counterfactual of 34.3 percent, for a gap of -7.5 percentage points. For Medicaid or means-tested public coverage, the observed post-period value was 54.9 percent, compared with an ASCM counterfactual of 43.1 percent, for a gap of +11.8 percentage points. For any public coverage, the observed post-period value was 55.5 percent, compared with an ASCM counterfactual of 43.5 percent, for a gap of +12.1 percentage points. For private coverage, the observed post-period value was 20.3 percent, compared with an ASCM counterfactual of 23.1 percent, for a gap of -2.8 percentage points.

The direction of the Medicaid and uninsurance estimates is consistent with the expected first-stage effect of expanding full-benefit coverage. The private-coverage result is secondary and should be interpreted cautiously because survey coverage categories can overlap and because private coverage may respond to composition, labor-market, or marketplace changes not fully captured by the first-pass donor pool.

Discussion

California shows a directionally promising coverage response, with lower uninsurance and higher Medicaid coverage in the exposed ACS target cell. In practical terms, the results suggest that the ACS can detect coverage changes in the low-income noncitizen target cell soon after implementation, at least in this first-pass synthetic-control screen.

The policy implications are straightforward but should remain measured. State-funded immigrant coverage expansions appear capable of moving public coverage and uninsurance in survey data, which matters for states considering expansions or retrenchments. However, the analysis does not identify undocumented residents directly and does not yet measure utilization, access, affordability, or safety-net financial outcomes.

Several limitations are important. First, ACS citizenship does not distinguish undocumented immigrants from lawful permanent residents, visa holders, refugees, or other noncitizen groups. Second, the post-period is short. Third, donor-pool contamination is possible because immigrant coverage policies are changing quickly across states. Fourth, pre-fit is very strong in several outcomes, which is encouraging for matching but also a reminder that many donor states and a short pre-period can produce fragile synthetic controls. Finally, the current analysis is an activation screen; publication-ready inference should add estimator comparisons and sensitivity checks before advancing causal language.

The next version should add SDID, generalized synthetic control, and matrix-completion estimates; audit state immigrant coverage policies year by year; test alternative age and income bands; and add access or utilization outcomes if public survey sources can support them.

Conclusion

California's Ages 26 through 49 Adult Full-Scope Medi-Cal Expansion is associated in this first-pass public-data screen with lower uninsurance and higher Medicaid coverage among low-income noncitizens ages 26-49. The result is promising enough to advance, but the manuscript should stay in a cautious working-paper frame until the donor policy audit and estimator-comparison pass are complete.

Sources

- California DHCS Adult Expansion: https://www.dhcs.ca.gov/services/medi-cal/eligibility/Pages/Adult-Expansion.aspx
- IPUMS ACS source extract documentation: `papers/noncitizen-coverage/data/scripts/01_extract_ac`

Tables

Table 1. Target-Population Coverage Summary

Group	Period	Years	Weighted population	Uninsured (%)	Medicaid (%)	Any public (%)	Private (%)
California	Pre-period	2018, 2019, 2020, 2021, 2022, 2023	961,502	35.8	42.2	42.8	23.1
California	Post-period	2024	895,554	26.8	54.9	55.5	20.3
Primary donor pool	Pre-period	2018, 2019, 2020, 2021, 2022, 2023	63,958	49.7	17.5	18.1	34
Primary donor pool	Post-period	2024	69,525	47.6	18.2	18.9	34.9

Notes: Cells are state-year means from ACS person-weighted state-year aggregates.

Table 2. Main ASCM Coverage Estimates

Outcome	Post years	Observed (%)	ASCM counterfactual (%)	Gap (pp)	Pre-RMSPE (pp)	Placebo rank p
Uninsured	2024	26.8	34.3	-7.5	0	0.341
Any public coverage	2024	55.5	43.5	12.1	0.85	
Medicaid / means-tested public coverage	2024	54.9	43.1	11.8	0.49	
Private coverage	2024	20.3	23.1	-2.8	0	

Notes: Gaps are treated-state observed minus ridge-augmented synthetic-control counterfactual. Positive gaps indicate higher coverage; negative gaps indicate lower uninsurance or lower private coverage.

Table 3. Largest Outcome-Specific Donor Weights

Outcome	Top donor states
Uninsured	MA (15.9%); HI (14.2%); RI (7.8%); VT (4.4%); OH (4.3%); ND (4.1%); IN (3.4%); PA (3.1%)
Any public coverage	MA (57.6%); NE (33.6%); DE (4.1%); AK (2.9%); VT (1.7%)
Medicaid / means-tested public coverage	MA (60.2%); NM (20.2%); SD (13.9%); AK (2.3%); VT (2.2%); DE (1.1%)
Private coverage	LA (28.0%); TN (15.9%); NM (11.0%); NJ (8.9%); KY (8.2%); AZ (7.5%); SC (6.9%); TX (5.4%)

Notes: Only the eight largest donor weights are displayed for each outcome.

Appendix

Appendix Figure A1. Raw ACS Pre-Trends

See `data/output/pretrend_acs_coverage.png`.

Appendix Figures A2-A5. Treated and Synthetic Paths

See `analysis/figures/synth_path_uninsured.png`, `analysis/figures/synth_path_medicaid.png`, `analysis/figures/synth_path_any_public.png`, and `analysis/figures/synth_path_private_coverage.png`.

Appendix Figures A6-A9. Gap Paths

See `analysis/figures/gap_path_uninsured.png`, `analysis/figures/gap_path_medicaid.png`, `analysis/figures/gap_path_any_public.png`, and `analysis/figures/gap_path_private_coverage.png`.

Appendix Table A1. Donor Weights

See `analysis/tables/table3_top_donor_weights.md` and `analysis/tables/donor_weights_by_outcome.csv`.

Appendix Table A2. Placebo-in-Space Diagnostics

See `analysis/robustness/placebo_in_space_uninsured.csv`.